

Tracing a Loop by Hand

Write down what every variable holds after every pass. This is how you find a bug without a debugger.

The code

```
total = 0
for i = 1 to 5:
    total = total + i
print(total)
```

The question to ask at every line: what changed?

Trace the boundaries first. Does the loop run 5 times or 6?
Does it start at 0 or 1? That is where the bug usually is.

The trace table

i	total	what just happened
-	0	before the loop starts
1	1	0 + 1
2	3	1 + 2
3	6	3 + 3
4	10	6 + 4
5	15	10 + 5
15 is what gets printed		

If your trace and the program disagree, one of you is wrong — and it is always worth finding out which.